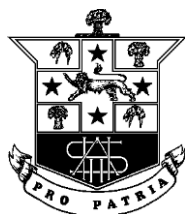


STUDENT NAME: _____

TEACHER: _____



HURLSTONE AGRICULTURAL
HIGH SCHOOL

2020

Year 11 Task 1

Mathematics Advanced

Examiners

- Mr. G. Huxley, Ms T. Tarannum, Ms L Yuen

General**Instructions**

- Reading time –
- Working time – 40 min (Submit by 4pm)
- Write using black pen
- NESA-approved calculators may be used
- A Reference sheet is provided (on Edmodo)
- In Questions 4 – 6, show relevant mathematical reasoning and/or calculations

Total marks:

30

Section I – 3 marks (pages 2 -3)

- Attempt Questions 1 – 3
- Allow about 4 minutes for this section

Section II – 27 marks (pages 4 – 9)

- Attempt Questions 4 – 6
- Allow about 36 minutes for this section

SECTION I

3 marks

Attempt Questions 1 – 3

Allow about 4 minutes for this section.

Use the multiple-choice answer sheet for Questions 1 – 3.

Question 1

Which of the following is the simplest form of $\frac{(m^{-2}n^3)^2}{m^{-4}n^{-1}}$?

- A. n^7
- B. n^4
- C. m^4n^7
- D. $\frac{n^7}{m^8}$

Question 2

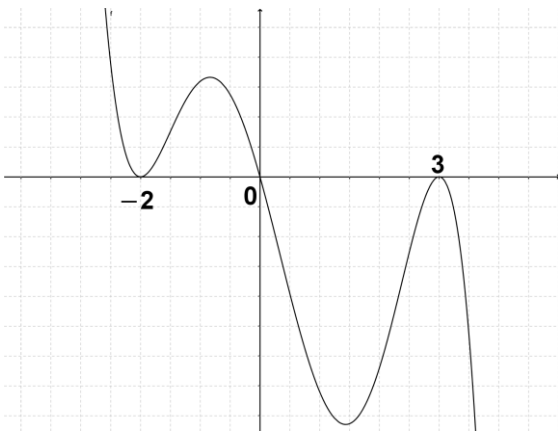
Which of the following straight lines is perpendicular to the line $3x + 6y - 4 = 0$?

- A. $6x + 3y - 4 = 0$
- B. $3x - 6y - 4 = 0$
- C. $6x - 3y - 4 = 0$
- D. $3x + 6y + 4 = 0$

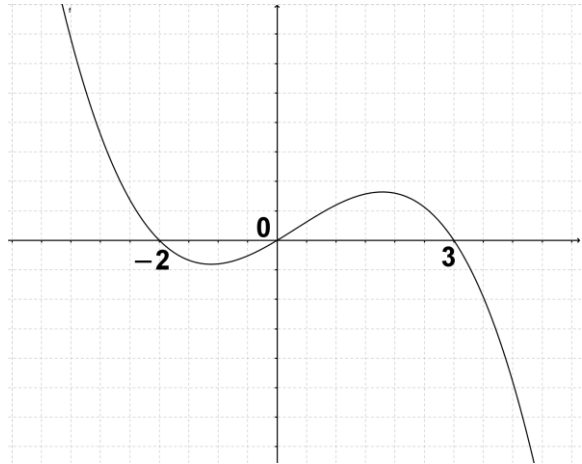
Question 3

Which of the following could be the graph of $y = x(x + 2)^2(x - 3)$?

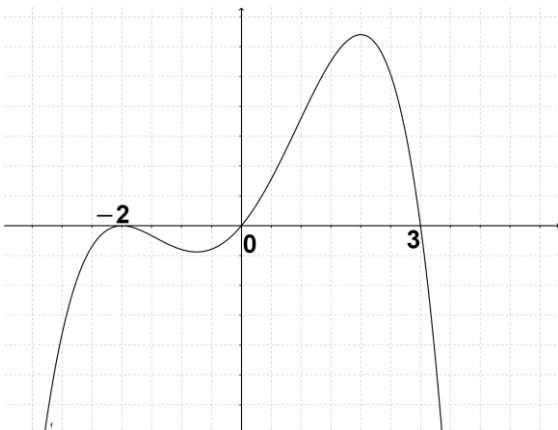
A.



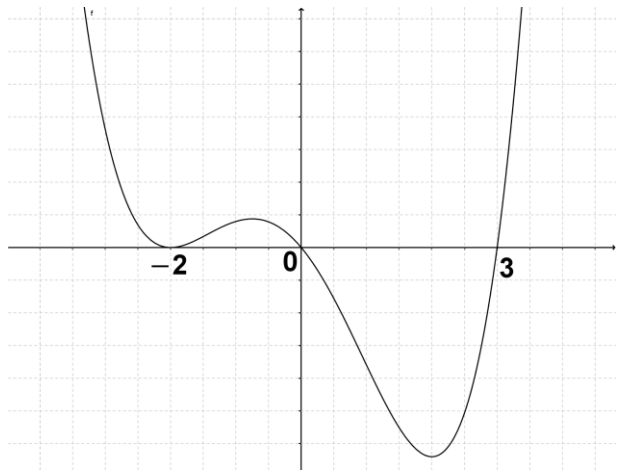
B.



C.



D.



End of Section I

Name: _____

SECTION II

27 marks

Attempt Questions 4 – 6

Allow about 36 minutes for this section.

In Questions 4 – 6, your responses should include relevant mathematical reasoning and/or calculations.

- **Write your name at the top of each page.** Answer the questions in the space provided. These spaces provide guidance for the expected length of response
- Extra writing space is provided at the back of this question booklet. If you used this space, clearly indicate which question and part you are answering.

Question 4 (9 marks)

Marks

- (a) Factorise the expression: $6a^2 - a - 2$ 2

- (b) Simplify fully: $\frac{a^3 + a^2b}{a^2 - b^2} \times \frac{a - b}{a}$ 2

Name: _____

- (c) Express the following with a rational denominator.

2

$$\frac{3}{\sqrt{3}-2} = \underline{\hspace{10cm}}$$

$$\underline{\hspace{10cm}}$$

$$\underline{\hspace{10cm}}$$

- (d) Use the method of completing the square to solve the following:
 $x^2 + 4x + 3 = 0$

3

End of Question 4

Name: _____

Question 5 (9 marks)

Marks

- (a) A function is defined by :

$$f(x) = \begin{cases} x^2 & \text{for } x < 0 \\ 2x & \text{for } 0 \leq x \leq 3 \\ 6 & \text{for } x > 3 \end{cases}$$

- (i) Find the value of $f(-3)$ **1**

- (ii) Draw a sketch of this function for $-5 \leq x \leq 5$ **2**

- (b) Given $f(x) = -x^2 + 2x$ and $g(x) = (x-1)$

- (i) Write a simplified expression for $f(g(x))$. **2**

- (ii) Write down the range of $f(g(x))$ in interval notation. **1**

Name: _____

- (c) (i) Write down the discriminant of $2x^2 + (k - 2)x + 8$, where k is a constant. **1**

- (ii) Hence, or otherwise, find the values of k for which the parabola

$$y = 2x^2 + kx + 9$$

does not intersect the line $y = 2x + 1$

2

End of Question 5

Name: _____

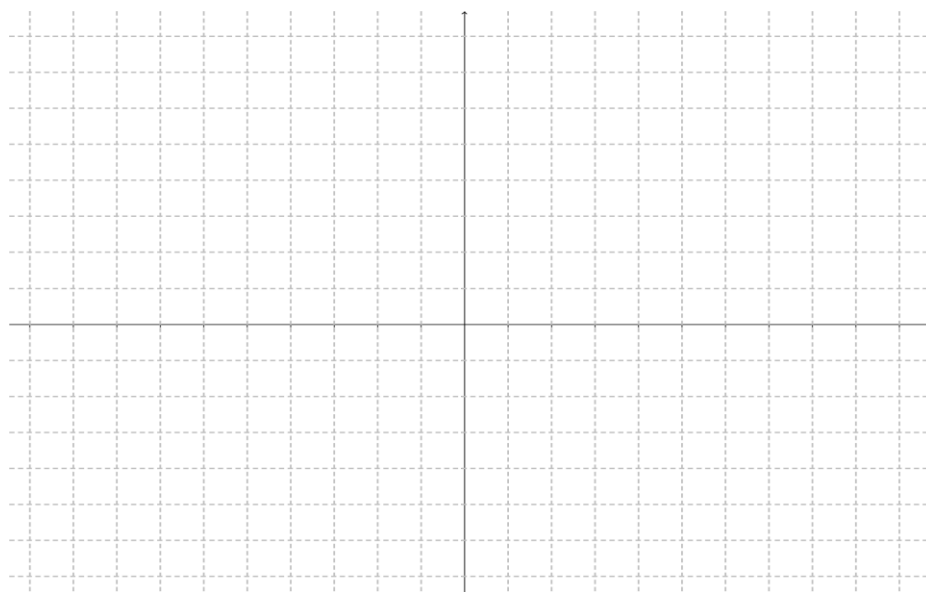
Question 6 (9 marks)

Marks

(a) Sketch the graph of $y=(x-2)^3+1$ on the number plane below.

Your sketch should include labels of the x and y intercepts.

2

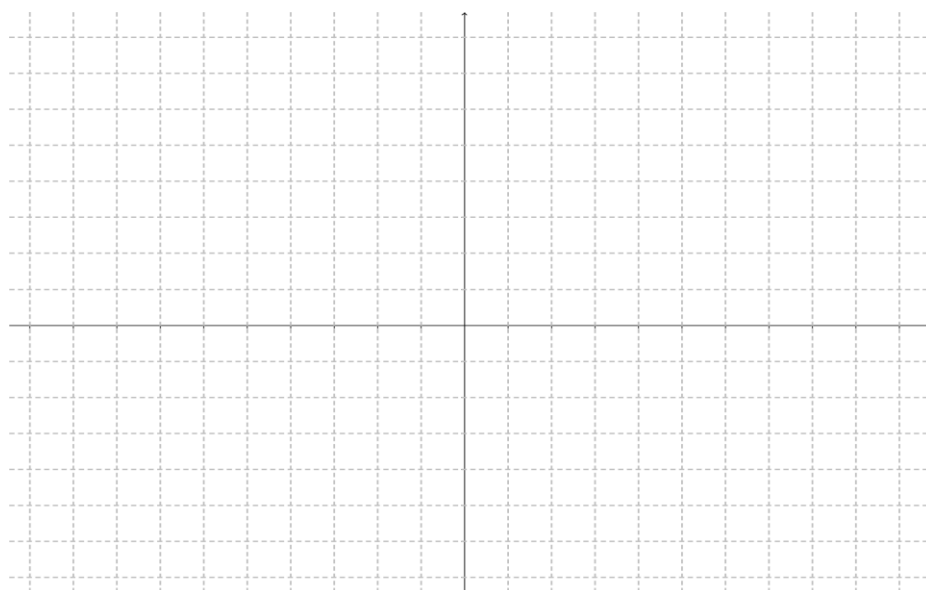


(b) (i) Sketch the graph of $x^2+(y-1)^2=4$ on the following number plane.

2

(ii) State the range of the graph in (i). _____

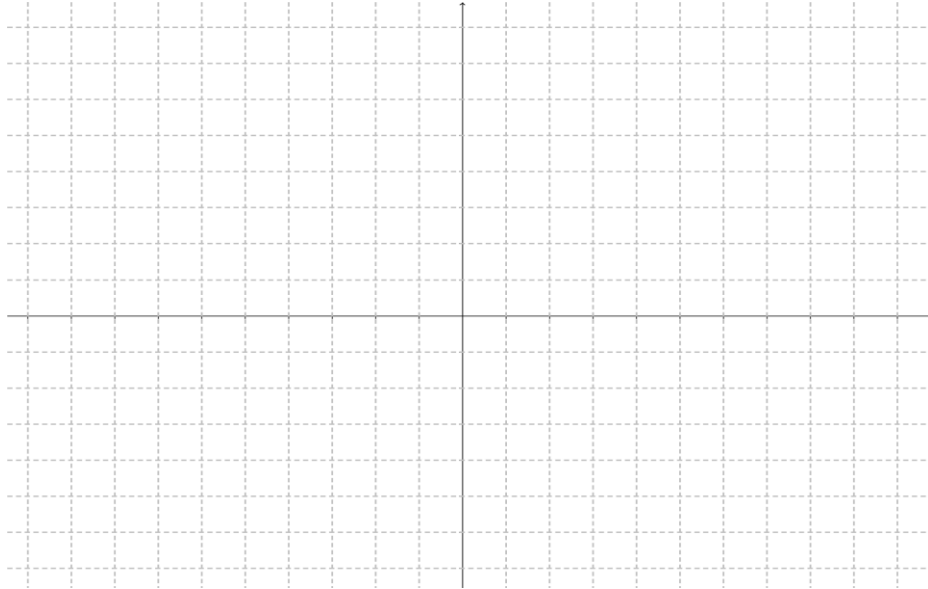
1



Name: _____

(c) (i) Sketch the graph of $y = \frac{9}{x}$ on the number plane below. **1**

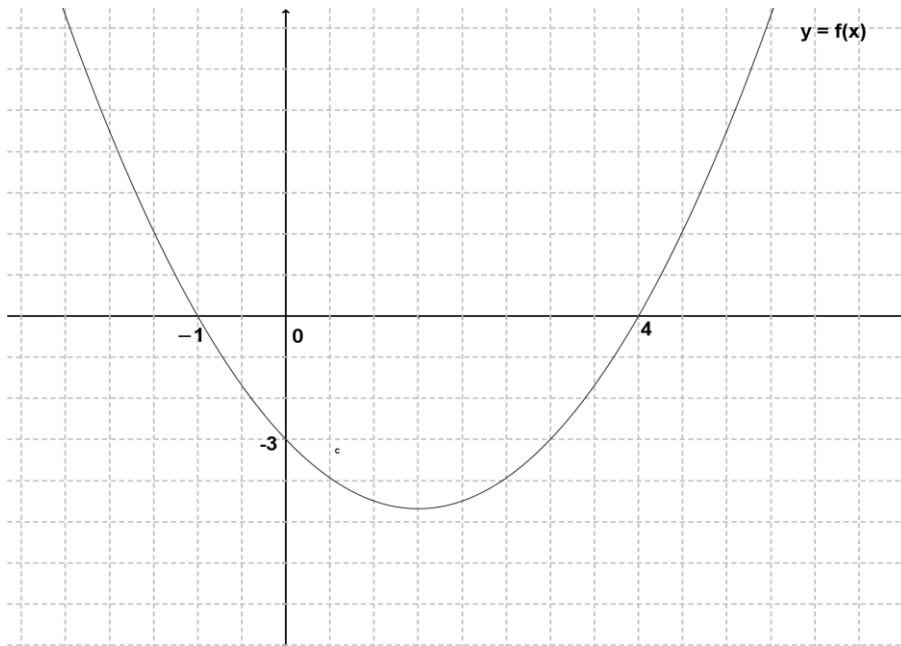
(ii) Label the point/s on the curve in (i) that are closest to the origin. **1**



(d) The graph of $y = f(x)$ is drawn below.

On the same number plane, draw the graph of $y = -f(x)$

Your sketch should be neat and accurate. **2**



End of Question 6

END OF TASK

Name: _____

Extra writing space

If you use this space, clearly indicate which part you are answering

Extra writing space with horizontal dashed lines for answers.



HURLSTONE AGRICULTURAL

HIGH SCHOOL

Student Name _____

Year 11

**Mathematics Advanced
2020 Assessment Task 1**

Teacher _____

MULTIPLE CHOICE ANSWER SHEET

--

Question

1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐

Question	Marks
MC	
4	
5	
6	
Total	30

All My Own Work Declaration:

I, _____ confirm that the enclosed material is all my own work.

I have not copied from any source apart from the research that I have done with relevant texts and technologies.



HURLSTONE AGRICULTURAL

HIGH SCHOOL

Year 11

**Mathematics Advanced
2020 Assessment Task 1**

MULTIPLE CHOICE ANSWER SHEET

Student Name DILLON

Teacher Solutions

Question

1. A ☒ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☒ D ☐
3. A ☐ B ☐ C ☐ D ☒

Question	Marks
MC	
4	
5	
6	
Total	30

All My Own Work Declaration:

I, _____ confirm that the enclosed material is all my own work.

I have not copied from any source apart from the research that I have done with relevant texts and technologies.

Name: _____

SECTION II

27 marks

Attempt Questions 4 – 6

Allow about 36 minutes for this section.

In Questions 4 – 6, your responses should include relevant mathematical reasoning and/or calculations.

- Write your name at the top of each page. Answer the questions in the space provided. These spaces provide guidance for the expected length of response
- Extra writing space is provided at the back of this question booklet. If you used this space, clearly indicate which question and part you are answering.

Question 4 (9 marks)

Marks

(a) Factorise the expression: $6a^2 - a - 2$

2

$$\begin{array}{l}
 \begin{array}{c} -4a \\ -a \end{array} \quad \begin{array}{c} -12a^2 \\ +3a \end{array} \quad \begin{array}{l} 6a^2 - 4a + 3a - 2 \\ = 2a(3a - 2) + 1(3a - 2) \\ = (3a - 2)(2a + 1) \end{array} \\
 \hline
 \text{2 marks - correct factorisation} \\
 \hline
 \text{1 mark - attempts to split middle term (or equivalent merit)}
 \end{array}$$

(b) Simplify fully:

$$\frac{a^3 + a^2b}{a^2 - b^2} \times \frac{a - b}{a}$$

2

$$\begin{array}{l}
 = \frac{a^2(a+b)}{(a-b)(a+b)} \times \frac{a-b}{a} \\
 \hline
 = a \\
 \hline
 \text{2 marks - correct solution} \\
 \hline
 \text{1 mark - factorisation of } a^3 + a^2b \text{ or } a^2 - b^2 \text{ (or equivalent merit)}
 \end{array}$$

Name: _____

- (c) Express the following with a rational denominator.

2

$$\frac{3}{\sqrt{3}-2} = \frac{3}{\sqrt{3}-2} \times \frac{\sqrt{3}+2}{\sqrt{3}+2}$$

2 marks — correct solution

$$= \frac{3(\sqrt{3}+2)}{3-4}$$

1 mark — multiplied by conjugate (or equivalent merit)

$$= \frac{3(\sqrt{3}+2)}{-1} \text{ or } -3(\sqrt{3}+2) \text{ or } -3\sqrt{3}-6$$

- (d) Use the method of completing the square to solve the following:

3

$$x^2 + 4x + 3 = 0$$

$$x^2 + 4x = -3$$

3 marks — correct answers (by correct method)

$$x^2 + 4x + (2)^2 = -3 + (2)^2$$

$$(x+2)^2 = 1$$

2 marks — generates perfect square
 ≠ take square root of both sides (or equivalent merit)

$$x+2 = \pm\sqrt{1} = \pm 1$$

$$\therefore x = -2 \pm 1 = -3 \text{ or } -1$$

1 mark — attempts to complete the square

End of Question 4

Question 5 (9 marks)

Marks

- (a) A function is defined by:

$$f(x) = \begin{cases} x^2 & \text{for } x < 0 \\ 2x & \text{for } 0 \leq x \leq 3 \\ 6 & \text{for } x > 3 \end{cases}$$

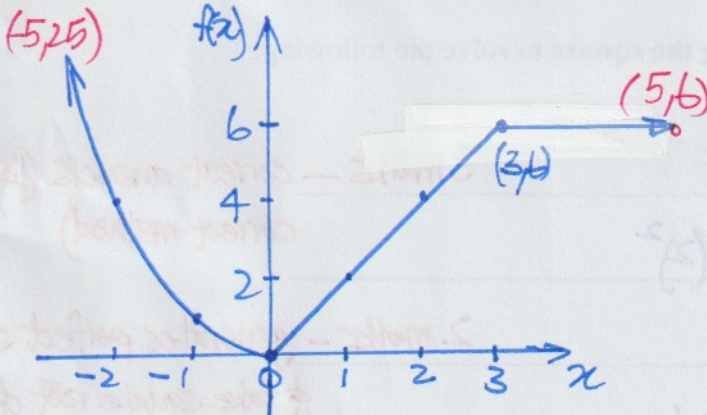
- (i) Find the value of
- $f(-3)$

1

$$f(-3) = (-3)^2 = 9$$

1 mark - correct answer

- (ii) Draw a sketch of this function for
- $-5 \leq x \leq 5$



* need to see / have implied the endpoints of domain.

2 marks - correct graph, showing / implied (3, 6) as the intersection *

1 mark - correct graph, without (3, 6) or 2/3 graphs correctly drawn.

- (b) Given
- $f(x) = -x^2 + 2x$
- and
- $g(x) = (x-1)$

- (i) Write a simplified expression for
- $f(g(x))$
- .

2

$$f(g(x)) = f(x-1) = -(x-1)^2 + 2(x-1)$$

2 marks - correct solution

$$= -(x^2 - 2x + 1) + 2x - 2$$

1 mark - initial substitution

$$= -x^2 + 4x - 3$$

$$= -(x^2 - 4x + 3) = -(x-3)(x-1)$$

for $f(x-1)$ (or equivalent merit)

- (ii) Write down the range of
- $f(g(x))$
- in interval notation.

$$= (3-x)(x-1) \text{ or } (x-3)(1-x)$$

$$f(g(x)) = -x^2 + 4x - 3 = -(x^2 - 4x + 3) = -(x-1)(x-3)$$

$$\text{Vertex occurs at } x = -\frac{b}{2a} = -\frac{4}{2 \cdot -1} = 2$$

1 mark - correct answer (in correct form)

$$\text{Vertex is } (2, 1).$$

$$\text{Range is } y \leq 1. \text{ In interval notation } (-\infty, 1]$$

Only need this.

Name: _____

- (c) (i) Write down the discriminant of $2x^2 + (k-2)x + 8$, where k is a constant. 1

$$\begin{aligned}\Delta &= b^2 - 4ac = (k-2)^2 - 4 \cdot 2 \cdot 8 \quad \text{156} \quad \text{1 mark - correct answer} \\ &= k^2 - 4k + 4 - 64 \\ &= k^2 - 4k - 60 \quad (k-10)(k+6)\end{aligned}$$

- (ii) Hence, or otherwise, find the values of k for which the parabola

$$y = 2x^2 + kx + 9$$

does not intersect the line $y = 2x + 1$

2

For the line & the parabola to intersect

$$2x^2 + kx + 9 = 2x + 1$$

$$2x^2 + kx - 2x + 8 = 0$$

$$\text{i.e. } 2x^2 + (k-2)x + 8 = 0 \quad \dots (1)$$

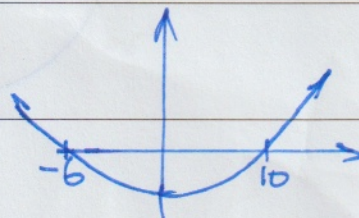
If the line & parabola do not intersect, Δ of (1) is such that $\Delta < 0$ (no real roots)

$$\text{i.e. } k^2 - 4k - 60 < 0$$

$$(k-10)(k+6) < 0$$

$$-6 < k < 10$$

From graph
(or alternate methods)
critical points, etc.



End of Question 5

2 marks — correct answer (with correct method shown)

1 mark — determines equation of quadratic (i.e. 1) or solves $\Delta < 0$ correctly

Name: _____

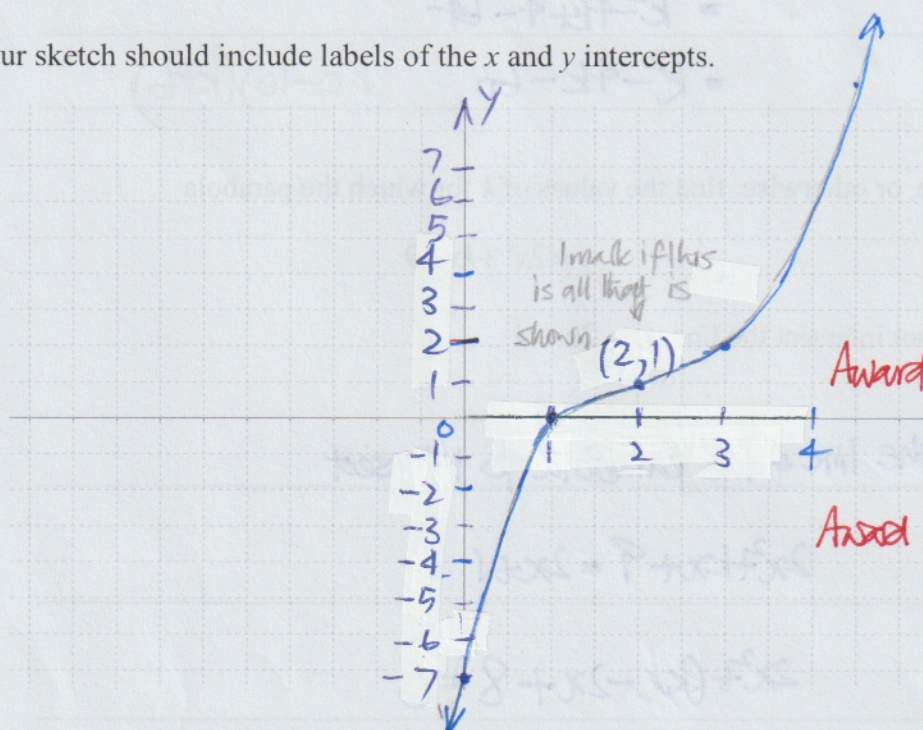
Question 6 (9 marks)

Marks

(a) Sketch the graph of $y=(x-2)^3+1$ on the number plane below.

Your sketch should include labels of the x and y intercepts.

2



Award 2 - correct graph with intercepts indicated

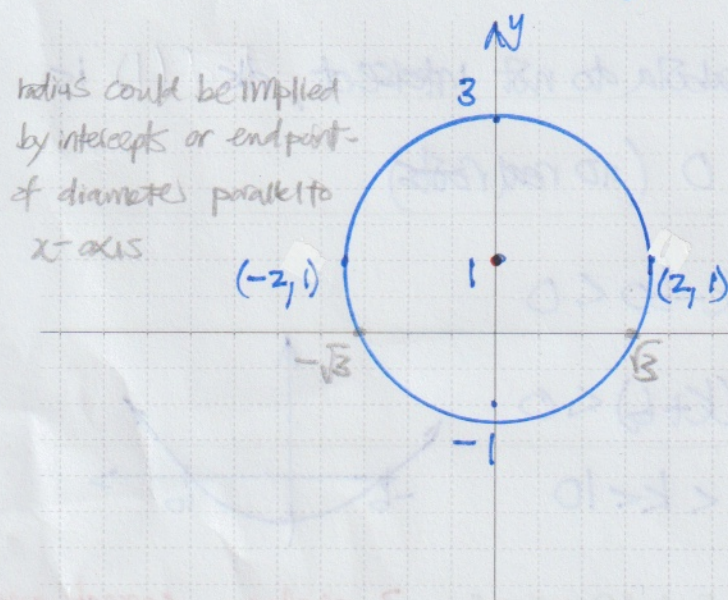
Award 1 - correct graph without both intercepts (or equivalent merit)

(b) (i) Sketch the graph of $x^2+(y-1)^2=4$ on the following number plane.

2

(ii) State the range of the graph in (i).

$-1 \leq y \leq 3$ or $[-1, 3]$ Award 1 + correct answer



Award 2 - correct graph (must be a circle) with both correct centre & correct radius

Award 1 - correct graph (must be a circle) but missing either radius or centre (or equivalent merit)

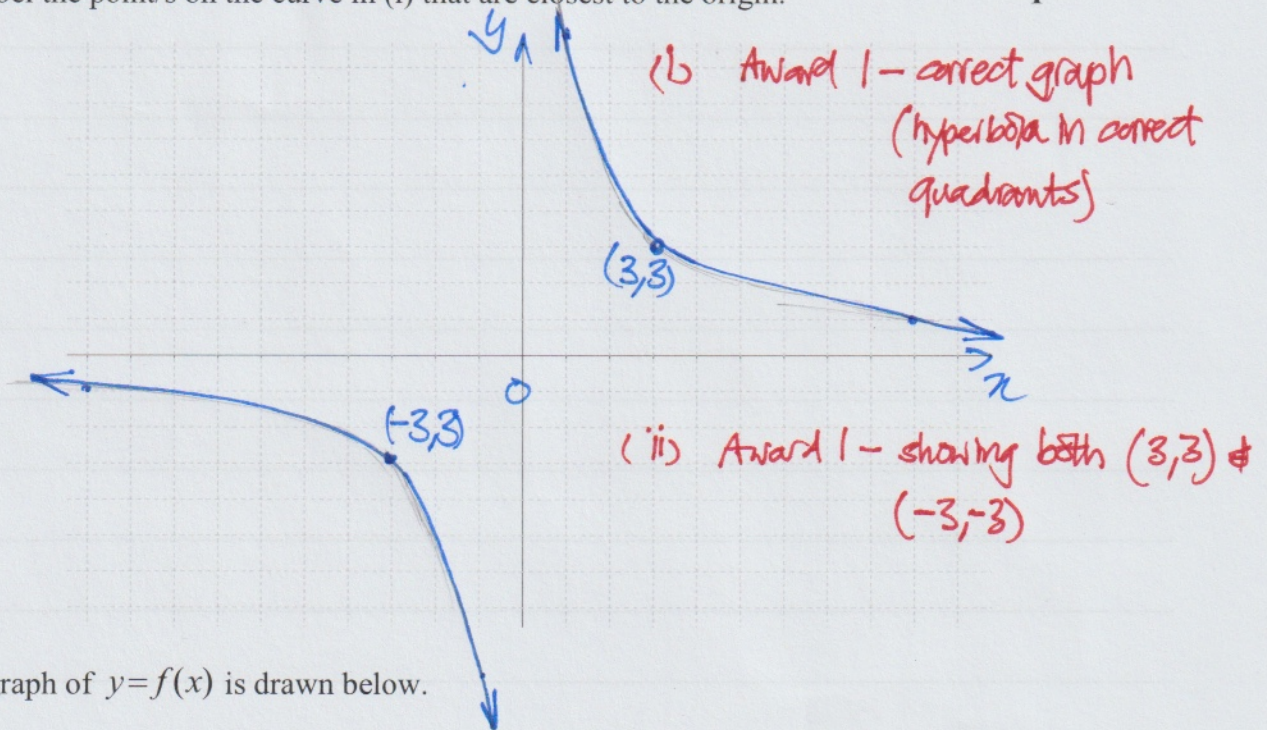
Name: _____

(c) (i) Sketch the graph of $y = \frac{9}{x}$ on the number plane below.

1

(ii) Label the point/s on the curve in (i) that are closest to the origin.

1

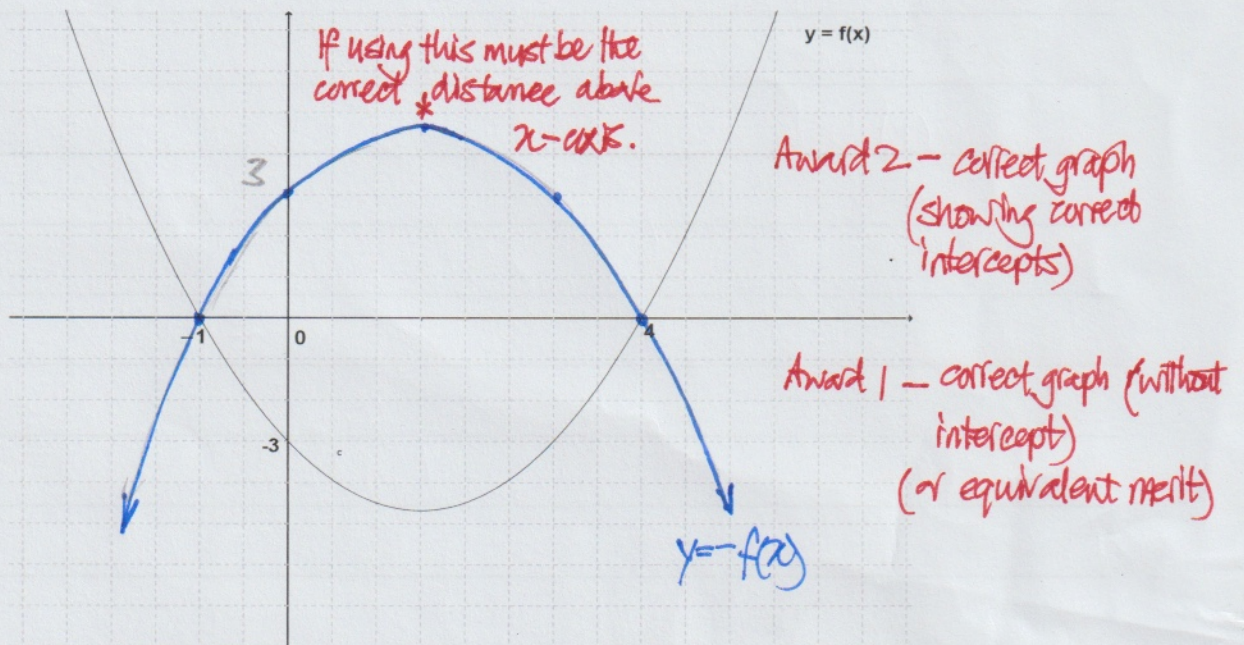


(d) The graph of $y = f(x)$ is drawn below.

On the same number plane, draw the graph of $y = -f(x)$

Your sketch should be neat and accurate.

2



End of Question 6

END OF TASK